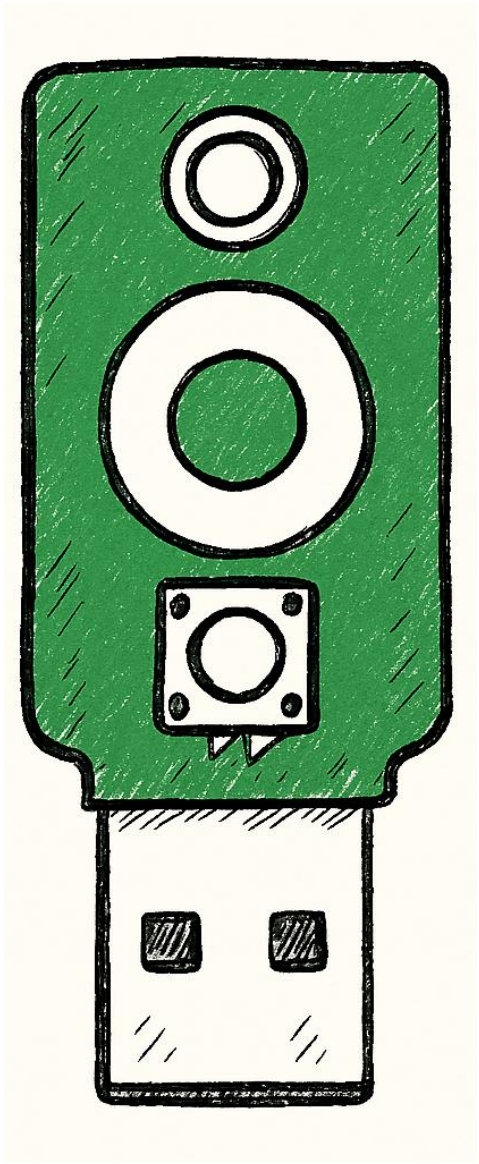


TouchAuth
A Biometric FIDO2 Key

A secure and user-friendly authentication solution using biometric technology

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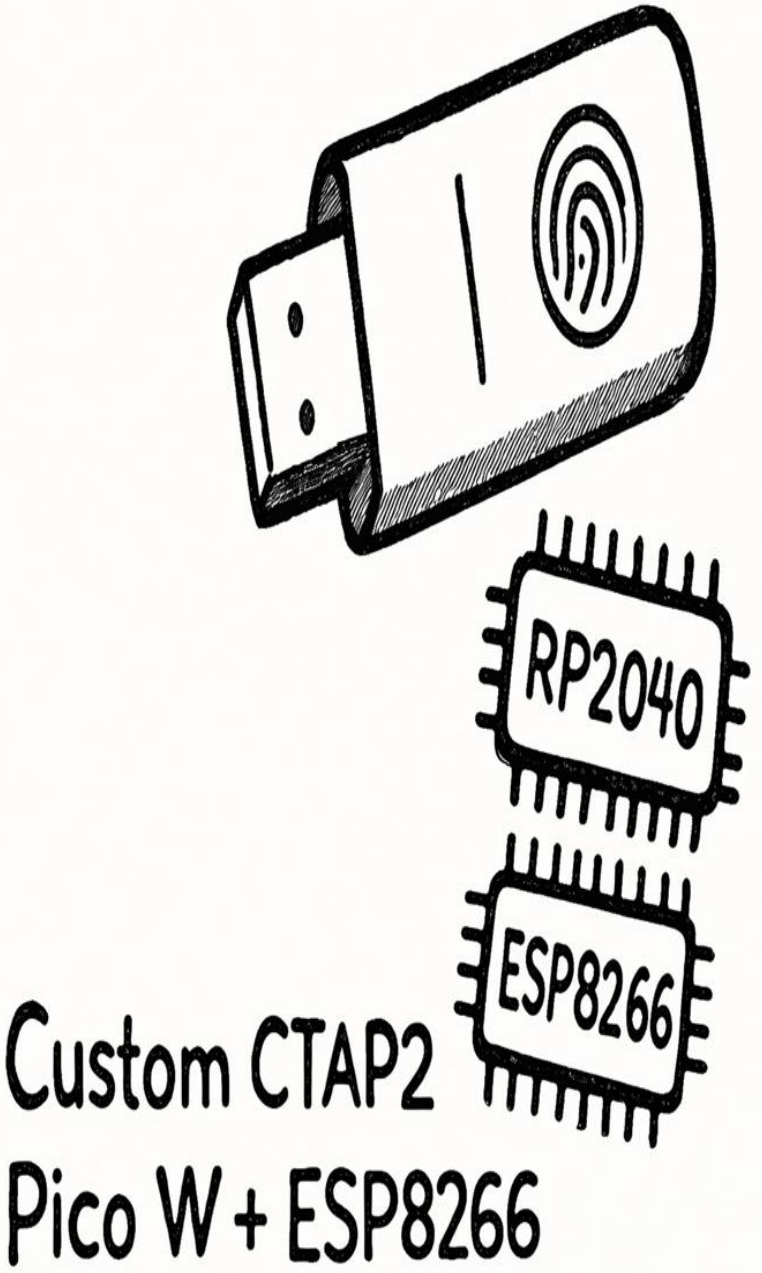
Abstract

The project titled “TouchAuth: A Biometric FIDO2 Key” presents a biometric-enabled hardware authentication system that integrates fingerprint verification with FIDO2-based security. The system uses an ESP8266 microcontroller and a fingerprint sensor to validate user identity before permitting authentication through a dedicated RP2040 Pico acting as a USB security key.Upon successful biometric verification, the ESP8266 generates a virtual user-presence signal to the Pico via GPIO. The Pico then performs secure cryptographic authentication with the connected computer using the FIDO2/WebAuthn standard.The architecture separates biometric processing from cryptographic operations, ensuring that sensitive data such as fingerprints and private keys remain isolated within their respective components.

Introduction

What is TouchAuth?

- Fingerprint-based biometric authentication.
- Hardware user presence verification.
- Implements custom-built FIDO2 firmware over USB HID.
- Integrates a match-on-sensor fingerprint module for local biometric verification.
- Enables in-browser authentication with no external apps or drivers.
- Supports Wi-Fi connectivity via ESP8266 for fingerprint registration.
- USB HID interface — no network risk.
- Open-source firmware for transparency.



Literature Survey (1)

Paper Name& Author(s)	Methodology Used	Merits	Demerits	Proposed System Benefit
Juan Lang, Alexei Czeskis, Dirk Balfanz, Marius Schilder, Sampath Srinivas Security Keys: Practical Cryptographic Second Factors for the Modern Web	Hardware-based authentication, challenge–response cryptography, attestation at registration, signature with counter at login, evaluated in Google's large-scale deployment.	Strong phishing resistance, easy to use, privacy-preserving.	Needs hardware token, limited browser support, higher cost.	Provides scalable, secure, passwordless authentication with reduced support burden.

Literature Survey (2)

Paper Name& Author(s)	Methodology Used	Merits	Demerits	Proposed System Benefit
Torque (Tareq) El Dandachi, Ashika Verma, Muhammad Abdullah U2F HID Implementation: Microprocessor to U2F Key	Hardware U2F key on Teensy 3.2, AES-128 encryption, APDU framing, HID communication, counter for anti-cloning, attestation certificate validation.	Implements FIDO2 U2F protocol on Teensy, strong security, WebAuthn compatibility.	Relies on hard-coded AES keys, limited hardware RNG, basic user presence via button.	Enables low-cost, reproducible U2F key with HID communication and WebAuthn support.

Literature Survey (3)

Paper Name& Author(s)	Methodology Used	Merits	Demerits	Proposed System Benefit
Zitao Zhang, Jacob Abbott, Sanchari Das, L.Jean Camp Building an Authentication Infrastructure: Designing a Two Factor Authentication Hardware Token	Survey of students with Yubikeys, Research Through Design prototyping of biometric hardware token, usability workshops, pilot Wizard-of-Oz testing, analysis of adoption barriers.	Affordable, user-friendly design, encourages engagement, strong two-factor security	Limited scalability, hardware dependency, potential usability gaps	Provides low-cost, accessible hardware token that improves adoption of secure authentication

Literature Survey (4)

Paper Name& Author(s)	Methodology Used	Merits	Demerits	Proposed System Benefit
Author from University of Gothenburg, Sweden (2019), Security Keys: An Empirical Study of FIDO2 Security Keys in Practice.	Compared FIDO2 authentication against traditional password logins.	High security, phishing resistance, improved login experience.	Usability gaps, user confusion, fallback to passwords.	Provides passwordless authentication with strong cryptographic security.

Literature Survey (5)

Paper Name& Author(s)	Methodology Used	Merits	Demerits	Proposed System Benefit
Wala'a M. AlOmari & Hesham Abusaimeh (2015), Modified USB Security Token for User Authentication.	OTP-based USB token authentication.	Dynamic passwords, portability, reduced password reliance.	Latency, limited hardware, token loss risk.	Secure identity verification via embedded OTP generation.

Existing System



YubiKey 5 NFC



YubiKey Bio (FIDO Edition)



Titan USB-A + NFC Key



Titan USB-C + NFC Key

Existing System

Disadvantages of existing system:

- Proprietary Design
- Expensive
- Setup & Recovery Complexity
- Hardware Limitations
- Lack of Customizability

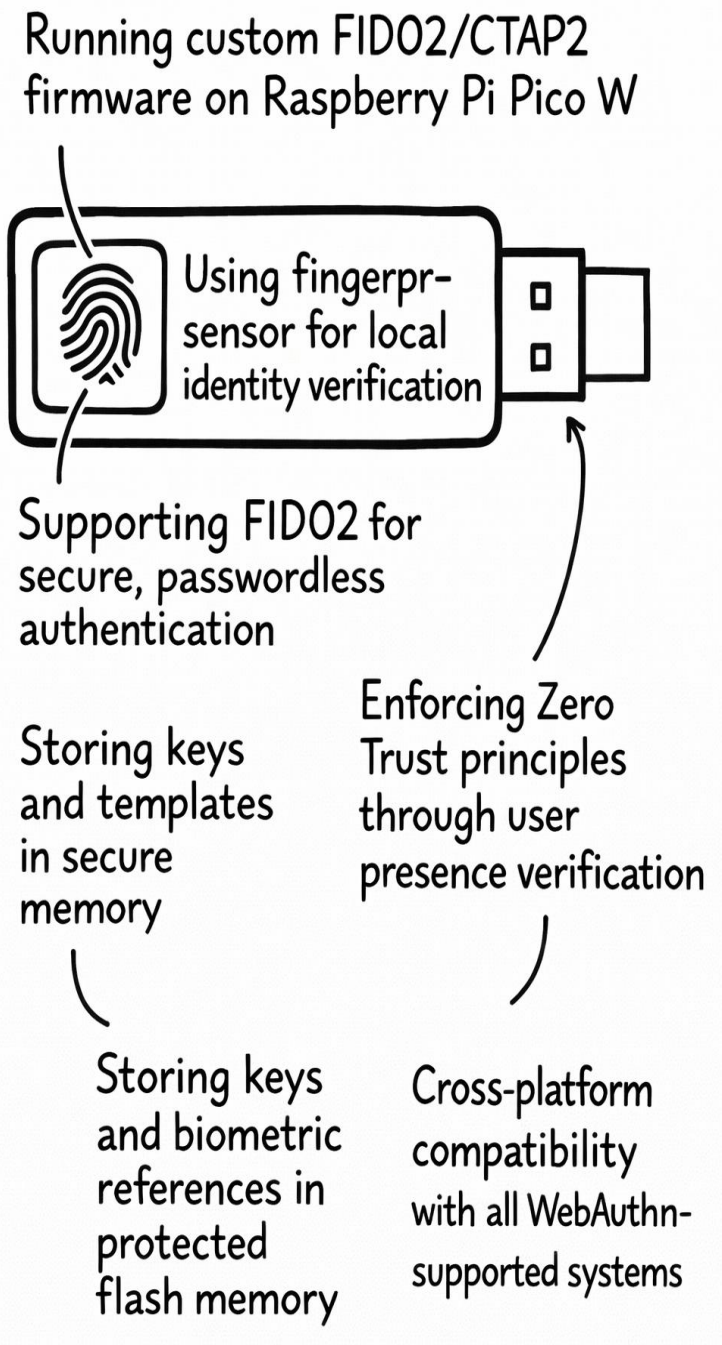


Problem Statement

Existing systems like YubiKey and SoloKeys offer strong public-key authentication; however, their high cost makes them less affordable, restricting large-scale adoption of hardware-based security solutions.

Proposed System

- Running custom FIDO2 firmware on RP2040.
- Using fingerprint sensor for local identity verification.
- Supporting FIDO2 for secure authentication.
- Connecting via USB for plug-and-play access.
- Storing keys and templates in secure memory.
- Enforcing Zero Trust with continuous identity checks.
- Cross-platform.
- Wi-Fi connectivity via ESP8266 for fingerprint registration.



Advantages

Key advantages of our project are:

- Hardware-Isolated Authentication.
- Provides strong multi-factor authentication.
- Seamless browser compatibility.
- No additional software installation required.
- Fast and user-friendly authentication process.
- Physically controlled user presence.
- Privacy-Preserving Authentication.
- Cross-Platform Compatibility.

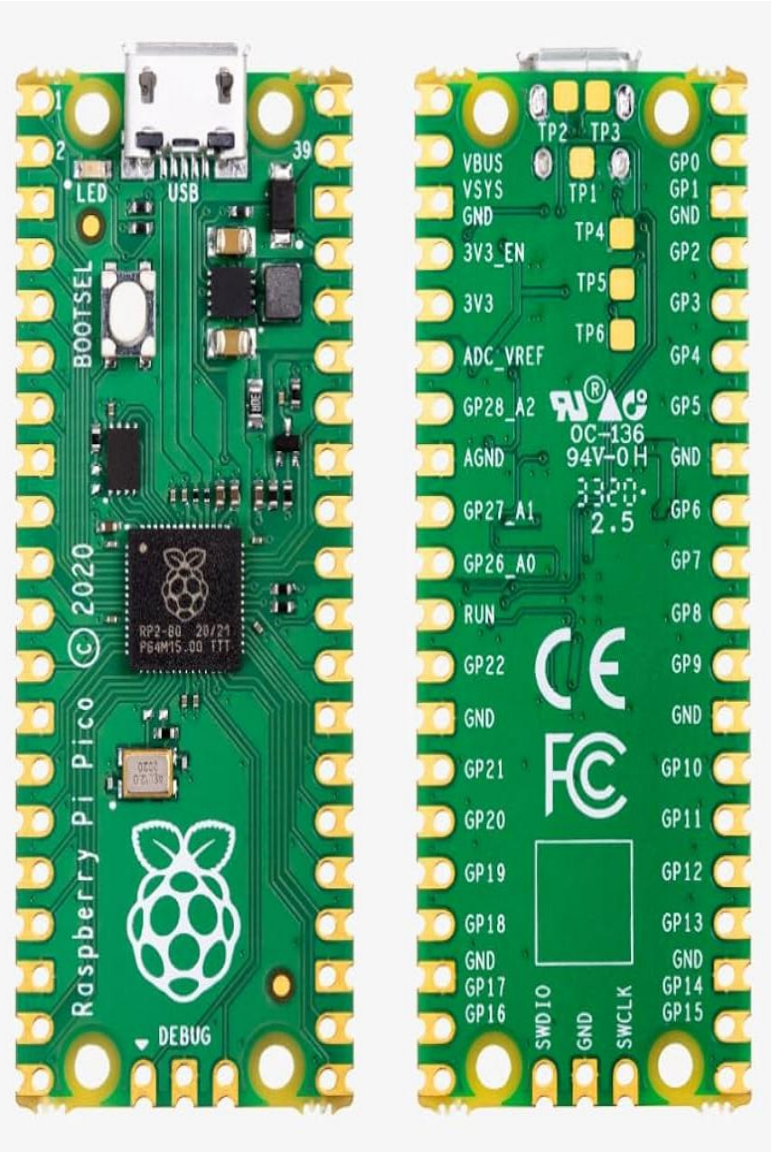


System Requirements

Hardware:

Raspberry Pi Pico W– RP2040 Microcontroller

- Core: Dual-core ARM Cortex-M0+ (32-bit).
- Clock Speed: Up to 133 MHz.
- Flash Memory: 2 MB onboard QSPI flash.
- SRAM: 264 KB on-chip SRAM.
- EEPROM: None (external storage optional).
- Random Number Generation: Software-based RNG.
- USB: USB 1.1 controller with device & host support (HID capable).
- Wireless: Integrated 2.4 GHz Wi-Fi (Pico W variant).
- Voltage: 1.8 V – 3.3 V I/O (VSYS up to 5.5 V input).

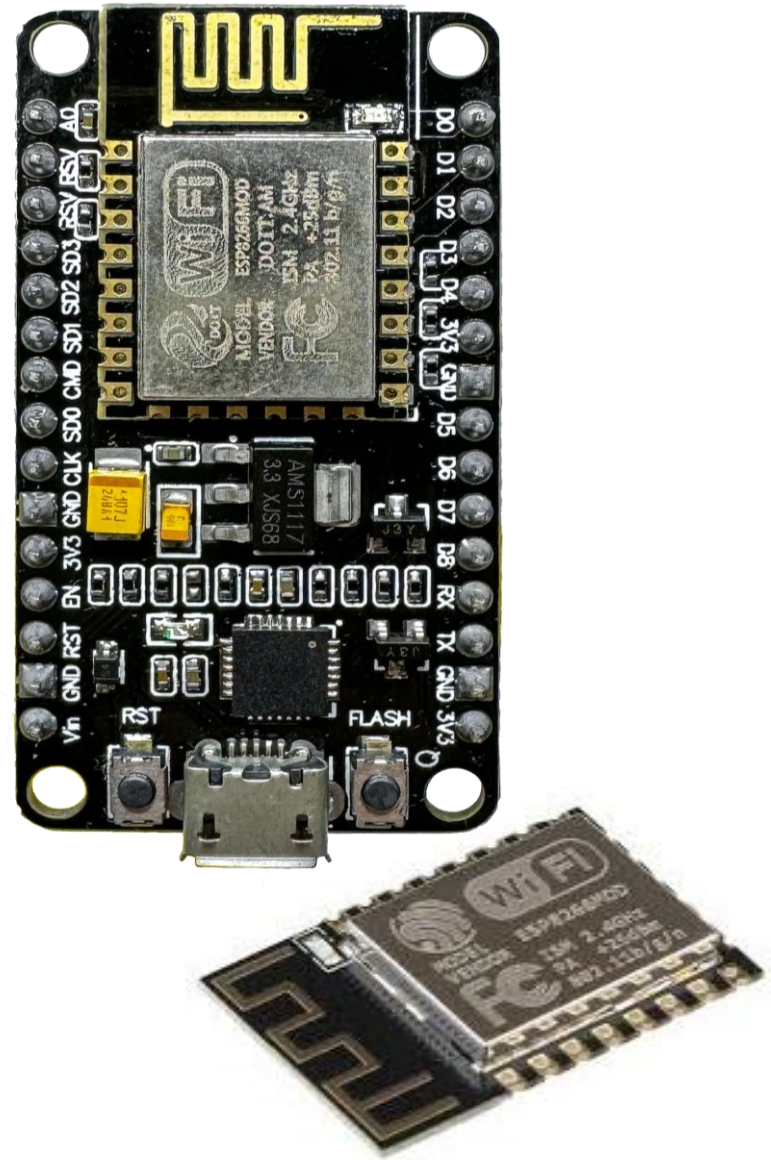


System Requirements

Hardware:

ESP8266 Microcontroller

- Core: Tensilica L106 (32-bit).
- Clock Speed: 80 MHz (up to 160 MHz overclocked).
- Flash Memory: 512 KB – 4 MB (board dependent).
- SRAM: ~160 KB on-chip SRAM.
- EEPROM: None (emulated in flash).
- Random Number Generation: Software-based RNG.
- Wi-Fi: Integrated 2.4 GHz 802.11 b/g/n (STA/AP mode).
- Digital I/O: ~11 GPIO pins (some reserved for flash).
- Analog Input: 1 channel (0 – 1 V range).
- UART: 1 Hardware UART (+ SoftwareSerial support).
- USB: Not available (requires external USB-to-Serial).
- Voltage: 3.0 V – 3.6 V operating (3.3 V typical).
- Current Consumption: ~70 – 200 mA during Wi-Fi transmission.

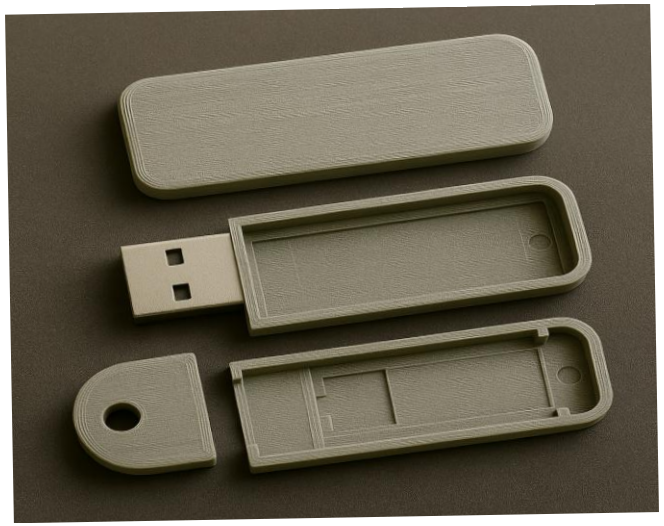


System Requirements

Hardware



R307 Fingerprint Sensor



A 3D-printed holder/case

System Requirements

Software

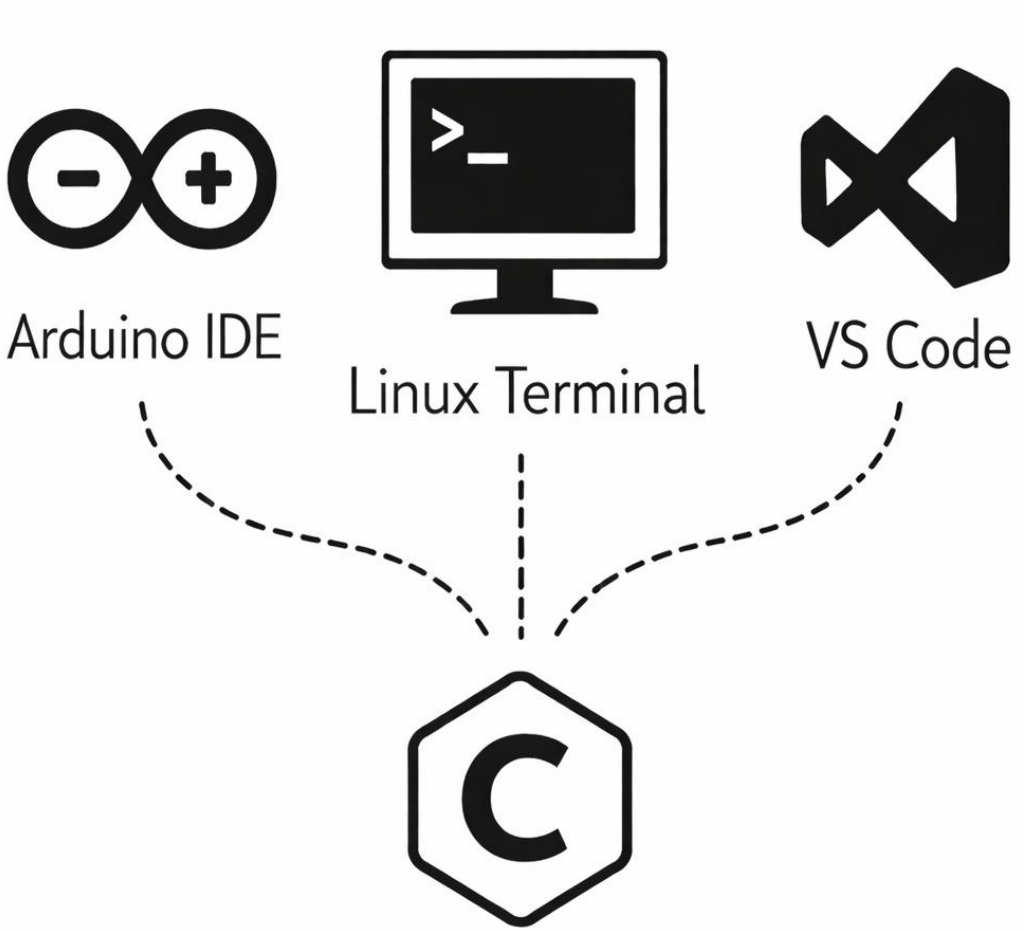
Programming Environment

Tools Used:

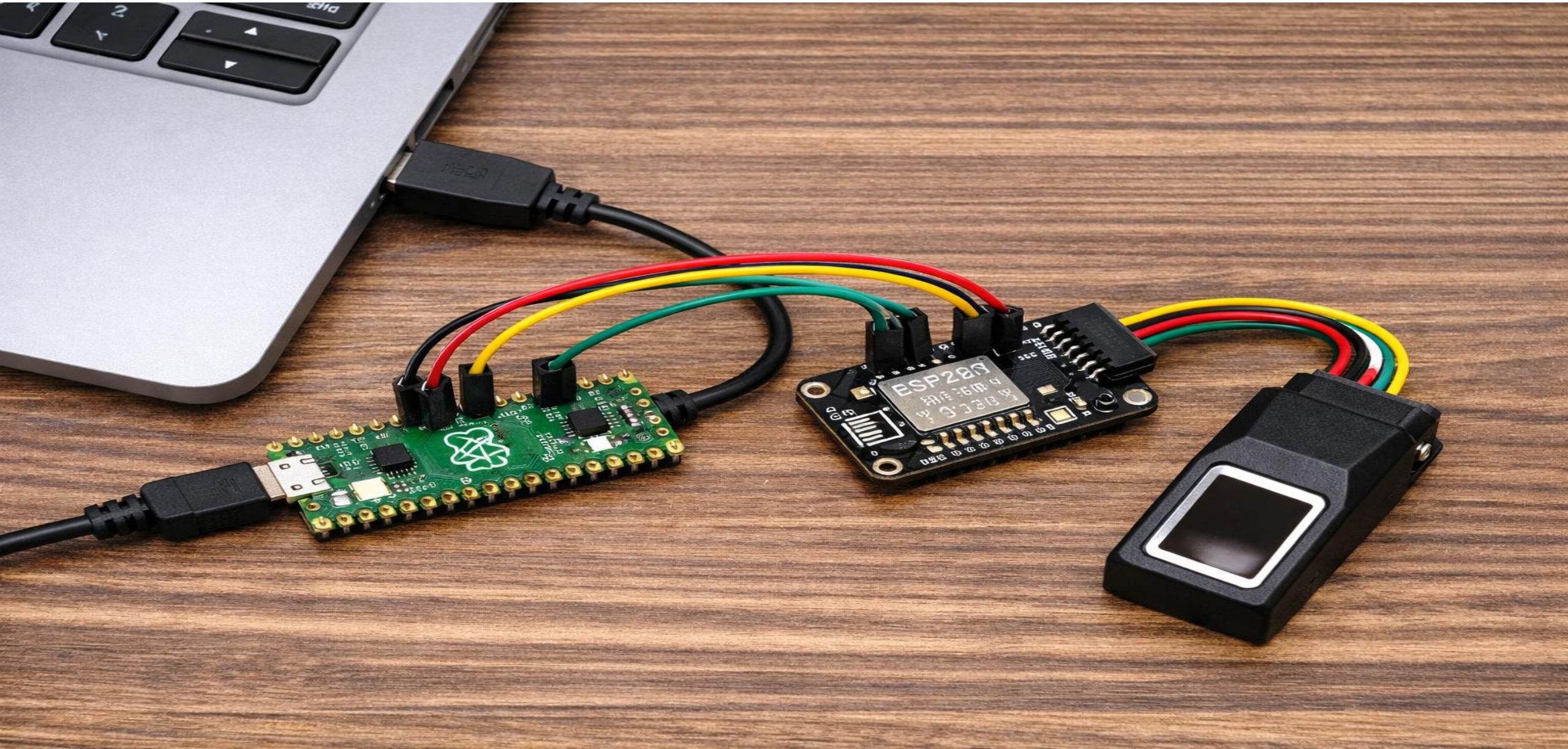
- Arduino IDE
- Linux Terminal
- VS Code

Firmware Languages:

- C

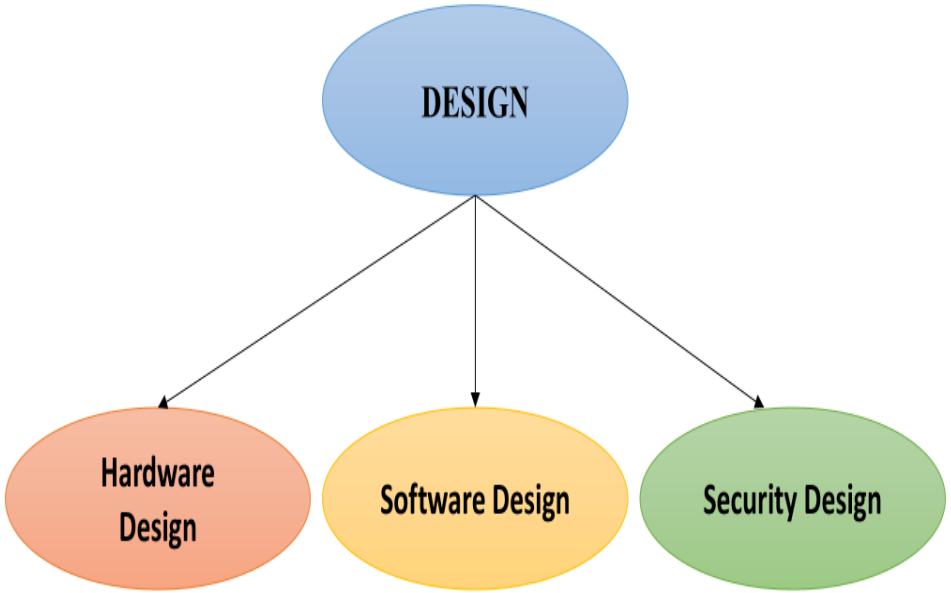


Design



System Design

Design Overview



The system is designed using the FIDO2 standard and divided into:

- Hardware Design
- Software Design
- Security Design

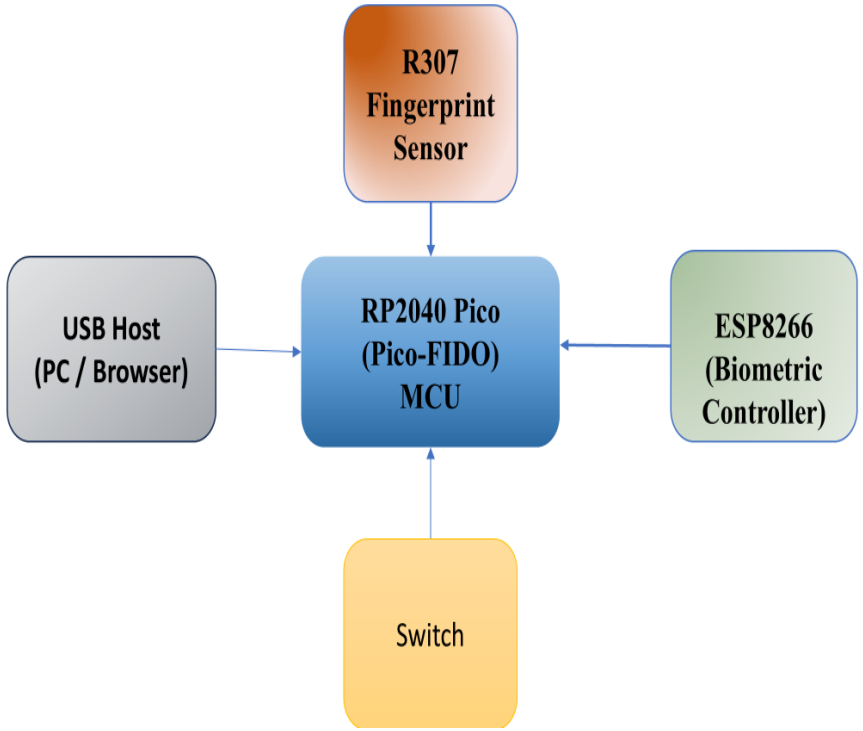
It combines:

- Fingerprint authentication
- Secure cryptographic key operations

Block Diagram

System Design

1. Hardware Design



Block Diagram

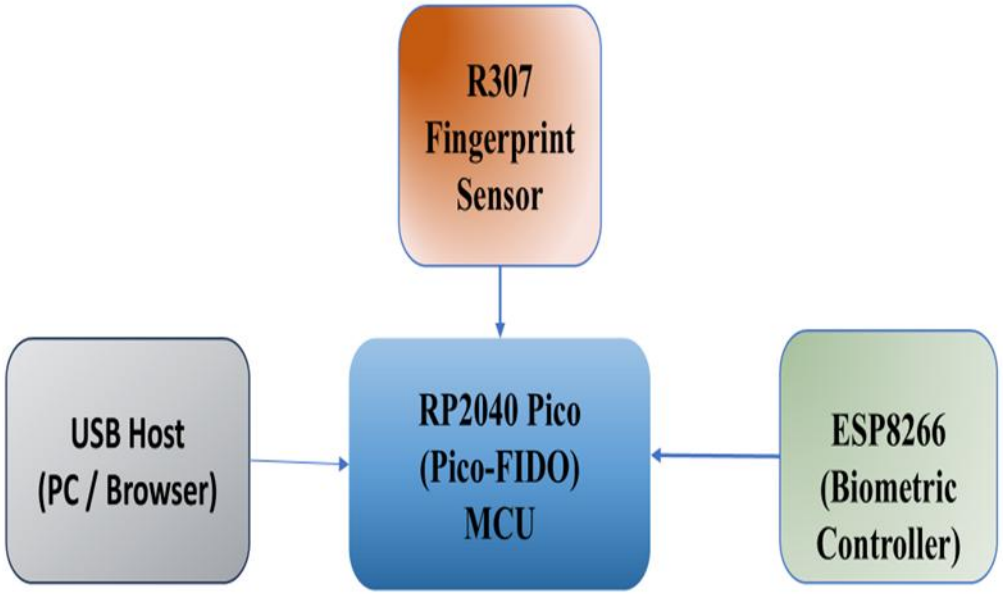
Hardware Architecture :

The hardware architecture follows a biometric-gated authenticator model, where fingerprint verification provides the user-presence signal for FIDO2 authentication:

- **RP2040 Pico (Pico-FIDO)**
  Operates as the USB security key implementing FIDO2/U2F protocols.
- **ESP8266 (Biometric Controller)**
  Handles fingerprint validation, user-presence signaling, and optional LED feedback.
- **R307 Fingerprint Sensor**
  Performs biometric capture and matching.
- **USB Host (PC / Browser)**
  Acts as the FIDO2 client communicating with the Pico.
- **SWITCH**
  To choose the mode.

System Design

1. Hardware Design



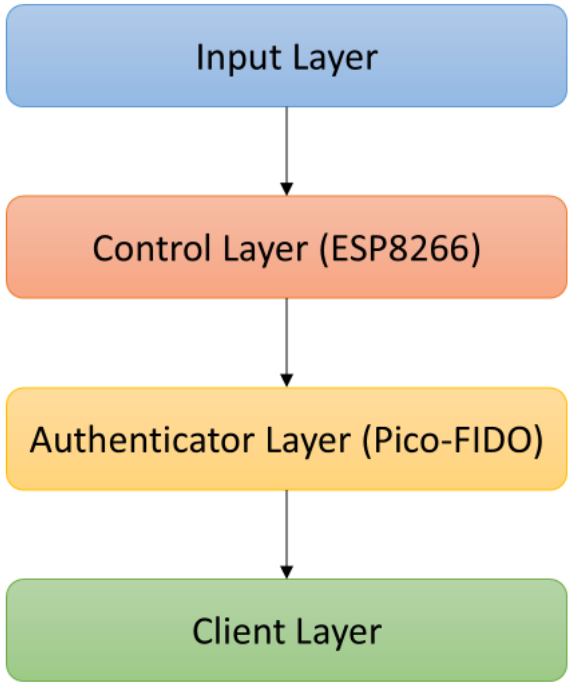
Block Diagram

Hardware Interconnections (Design Level)

Component	Interface	Purpose
Fingerprint Sensor → ESP8266	UART	Biometric verification
ESP8266 → Pico	GPIO	Virtual user-presence signal
Pico → PC	USB	FIDO2 authentication

System Design

2. Software Design



Block Diagram

Software Architecture:

The software design is layered and modular:

1. Input Layer

- Fingerprint capture.
- Sensor communication.

2. Control Layer (ESP8266)

- Biometric decision making.
- Authentication sequencing.
- Error handling.

3Authenticator Layer (Pico-FIDO)

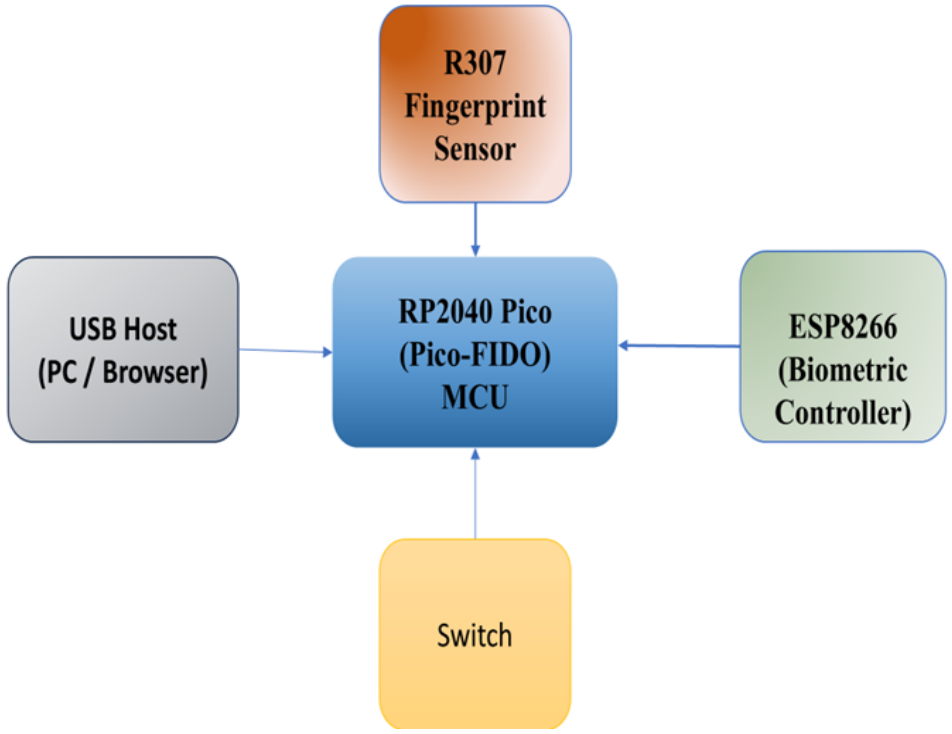
- USB stack (TinyUSB).
- CTAP2 / U2F protocols.
- Challenge–response handling.

4. Client Layer

- Browser WebAuthn API.
- OS-level security key support.

System Design

3. Security Design



Block Diagram

Authentication Model :

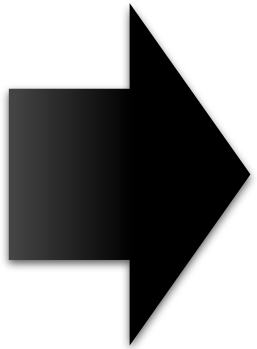
The system enforces multi-factor hardware authentication:

1. Something you have → Hardware device
2. Something you are → Fingerprint
3. Cryptographic proof → FIDO2 private keys

Key Management Design :

- Key pairs are generated during registration.
- **Private keys:**
  - Stored inside authenticator memory.
  - Never transmitted externally.
- **Public keys:**
  - Registered with relying party (server/browser).

Implementation

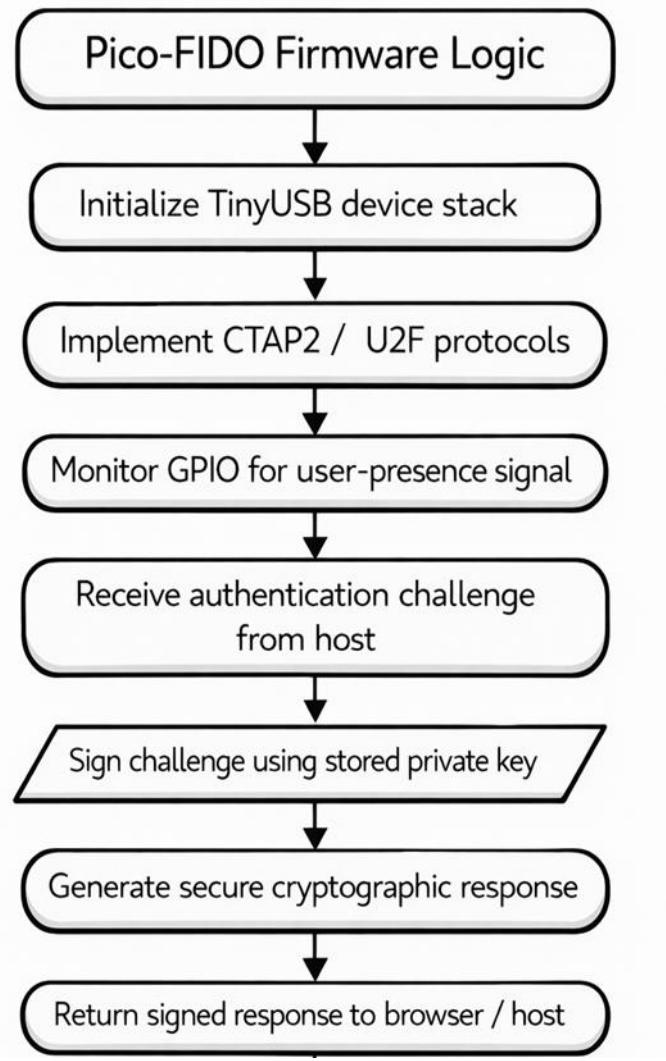


Pico-FIDO Firmware Logic:

- Initialize TinyUSB device stack.
- Implement CTAP2 / U2F authentication protocols.
- Monitor GPIO for user-presence signal.
- Receive authentication challenge from host.
- Sign challenge using stored private key.
- Generate secure cryptographic response.
- Return signed response to browser / host.

Key Principle

- Pico never processes fingerprint data.
- Biometric handling is isolated from cryptographic operations.

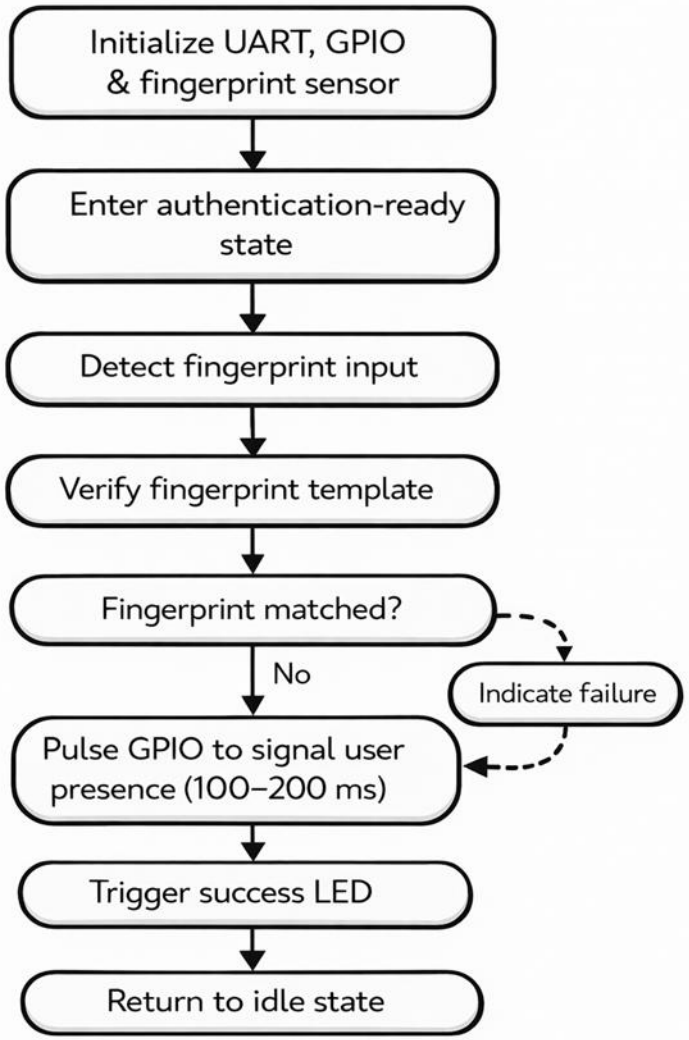


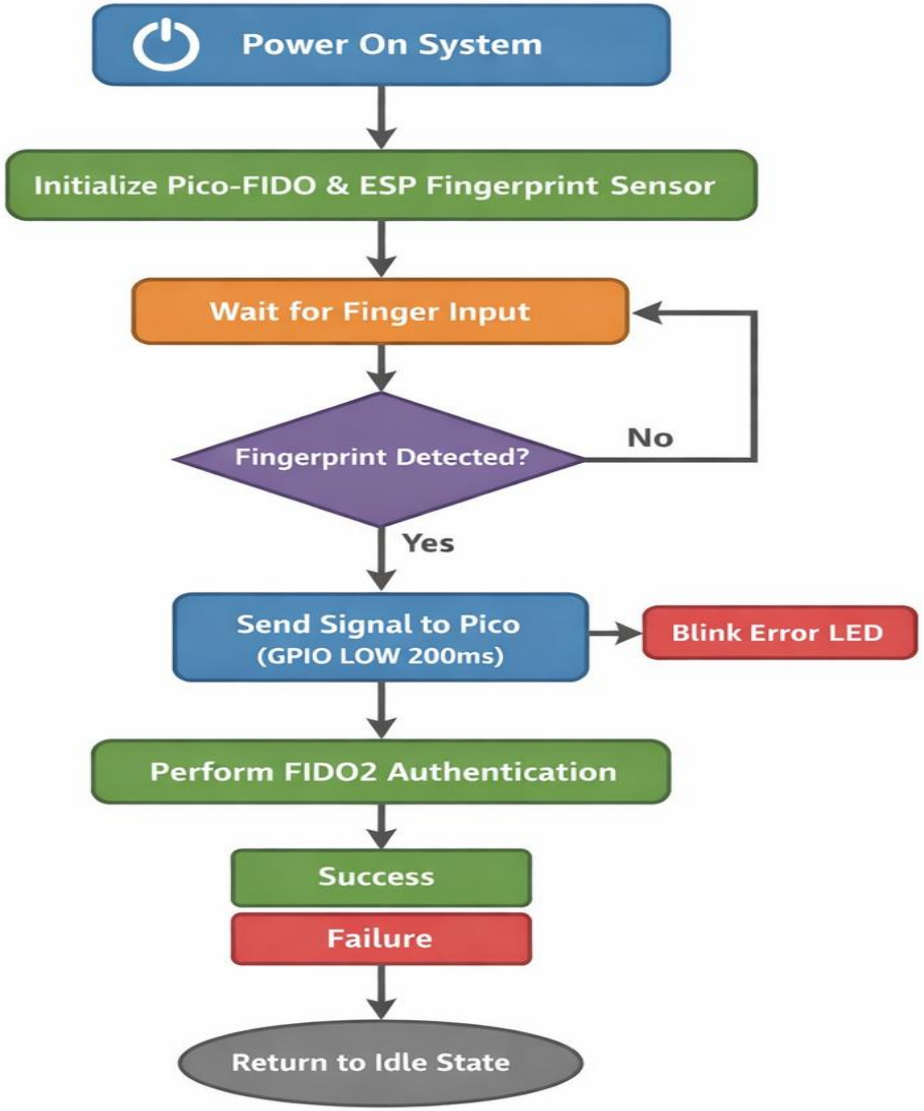
ESP8266 Firmware Logic:

- Initialize UART, GPIO, and fingerprint sensor.
- Enter authentication-ready state.
- Detect fingerprint input.
- Verify fingerprint template.
- If matched → Pulse GPIO (100–200 ms) to signal user presence.
- Trigger success LED; otherwise indicate failure.
- Return to idle state.

Key Principle

- ESP8266 performs biometric authorization only.
- Does not participate in cryptographic authentication.





Usage Diagram

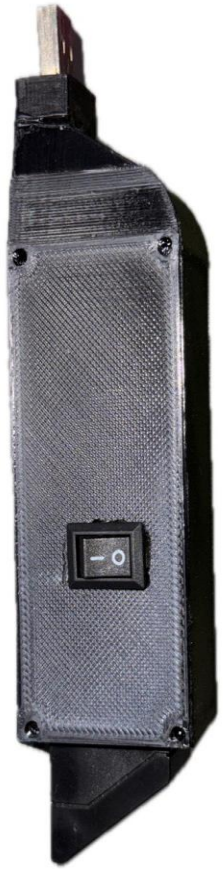
RESULT

TouchAuth Device



TouchAuth Device Plugged-In

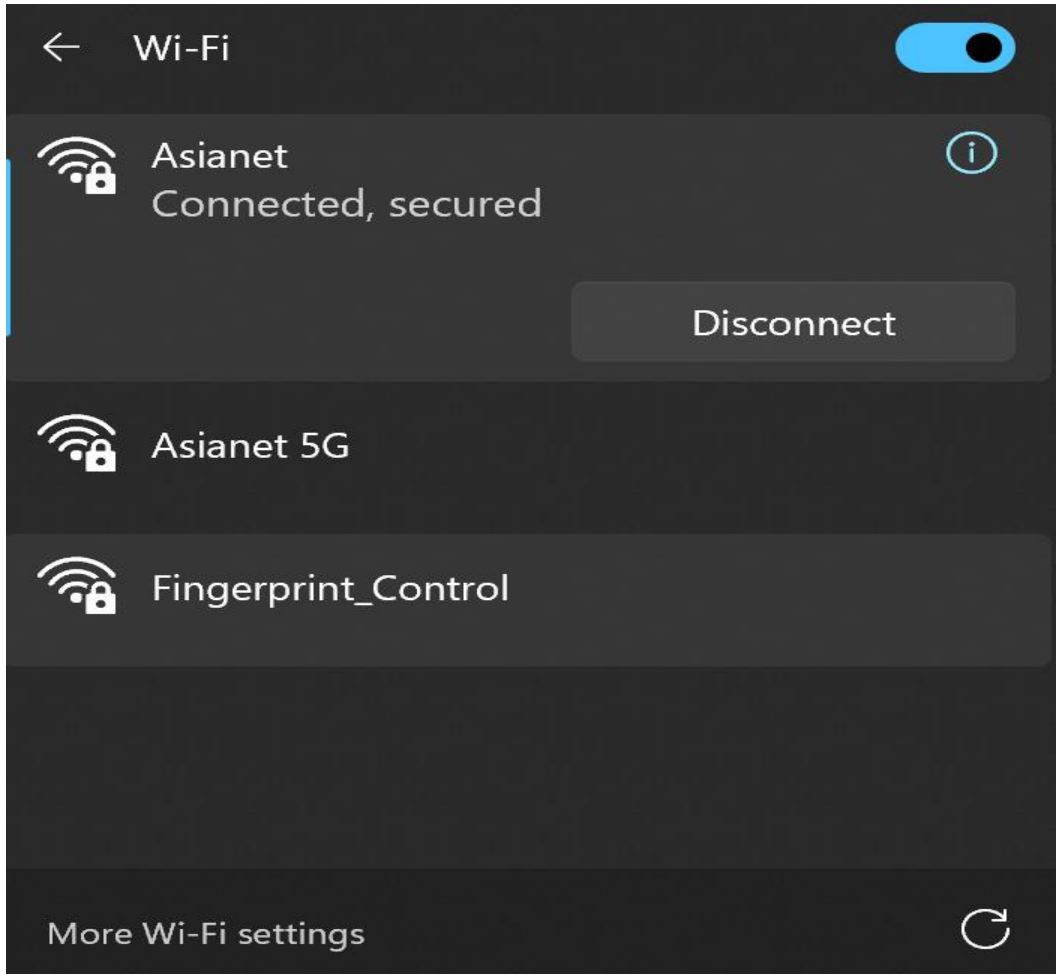




TouchAuth Device

RESULT

Mode 1: Fingerprint Registration



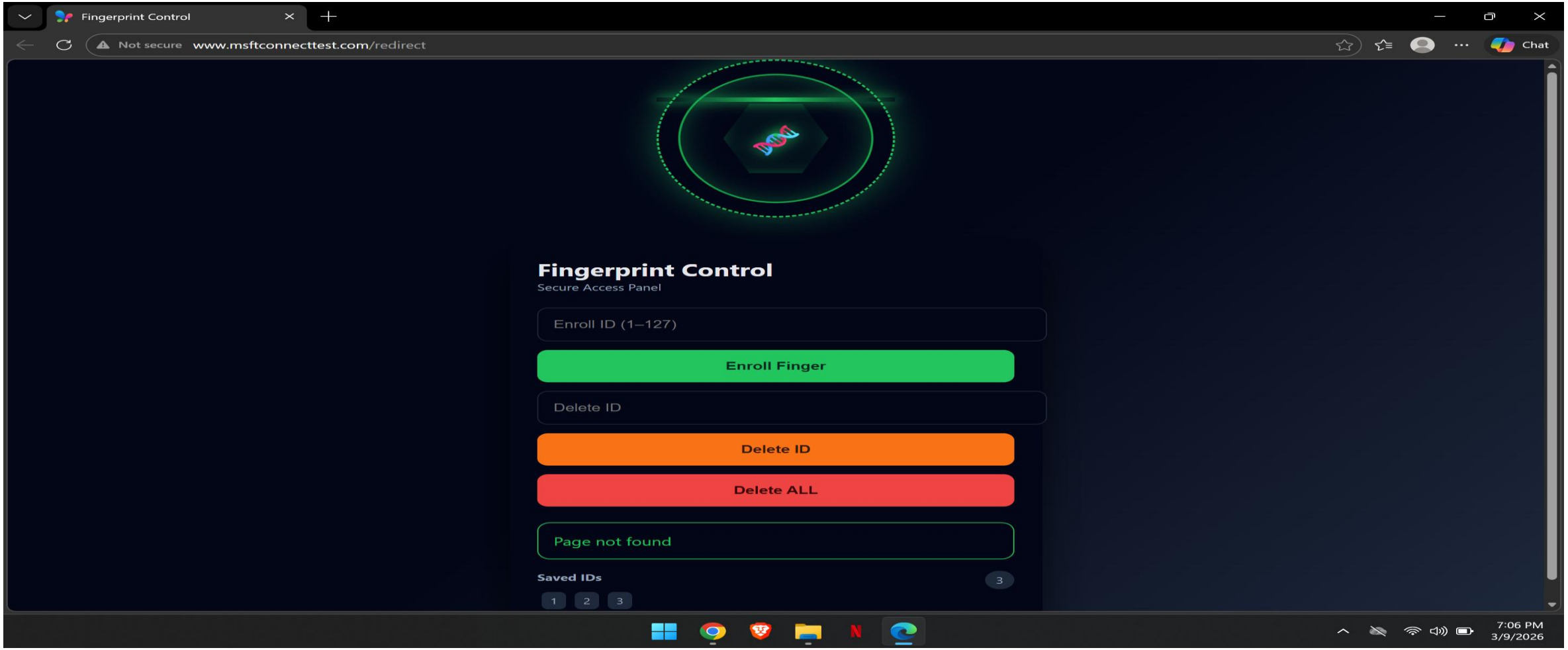
Device Detected By WIFI



Connected With WIFI Action Required

RESULT

Mode 1: Fingerprint Registration

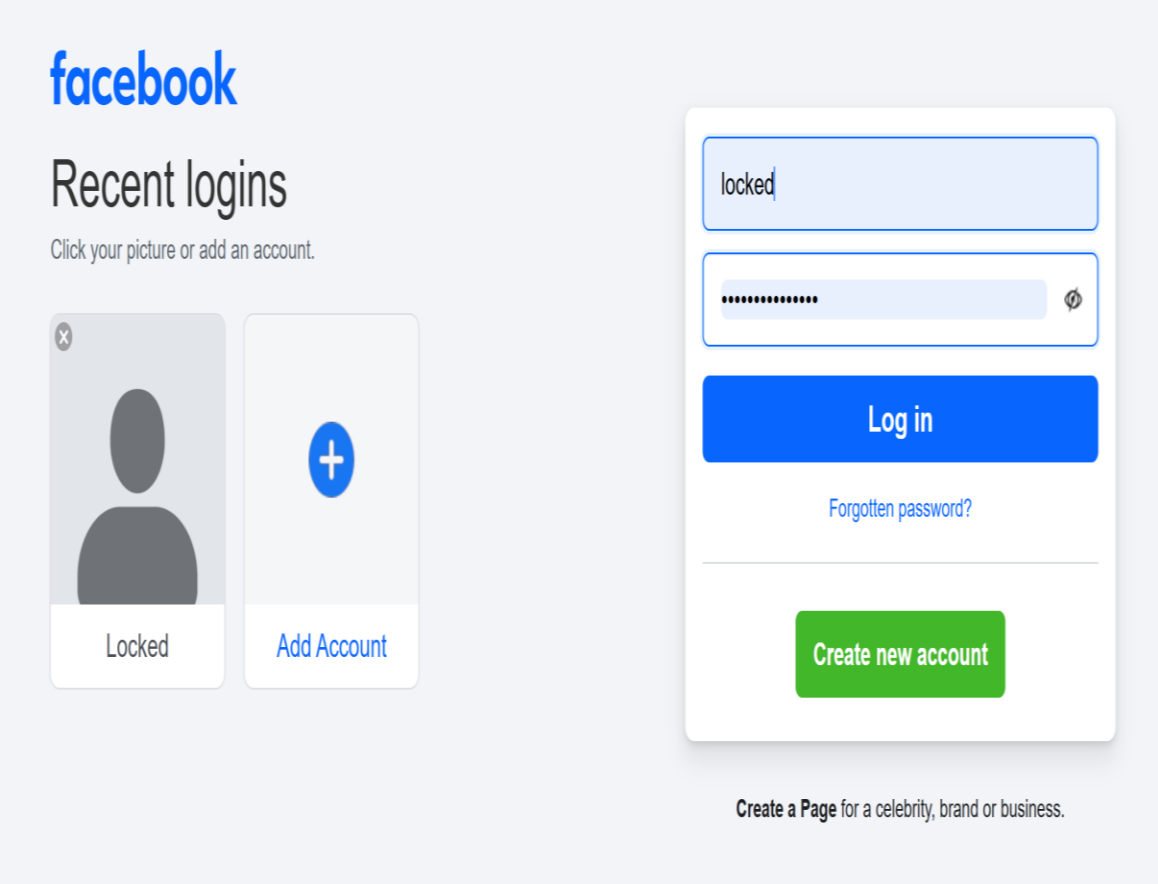


Web Page For Fingerprint Registration And Deletion

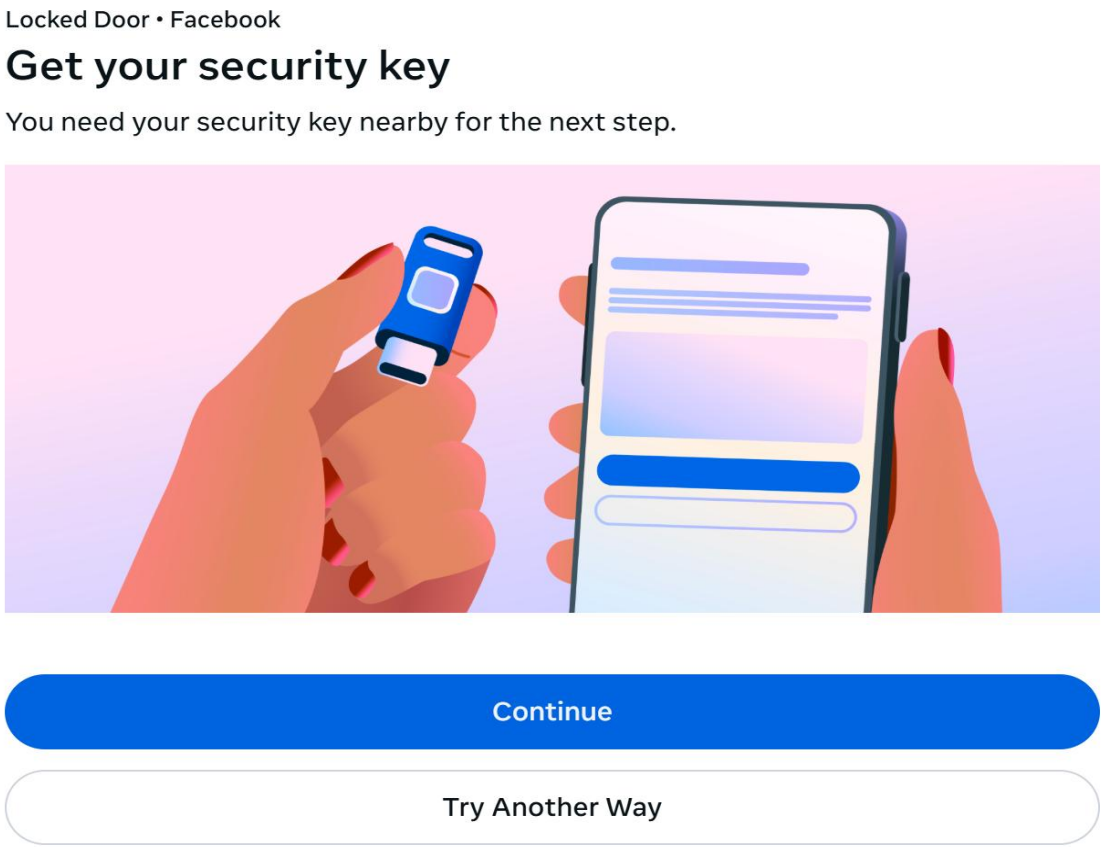
RESULT

Login Process (2-Factor Authenticaton)

Step 1
Username And Password Based Authentication



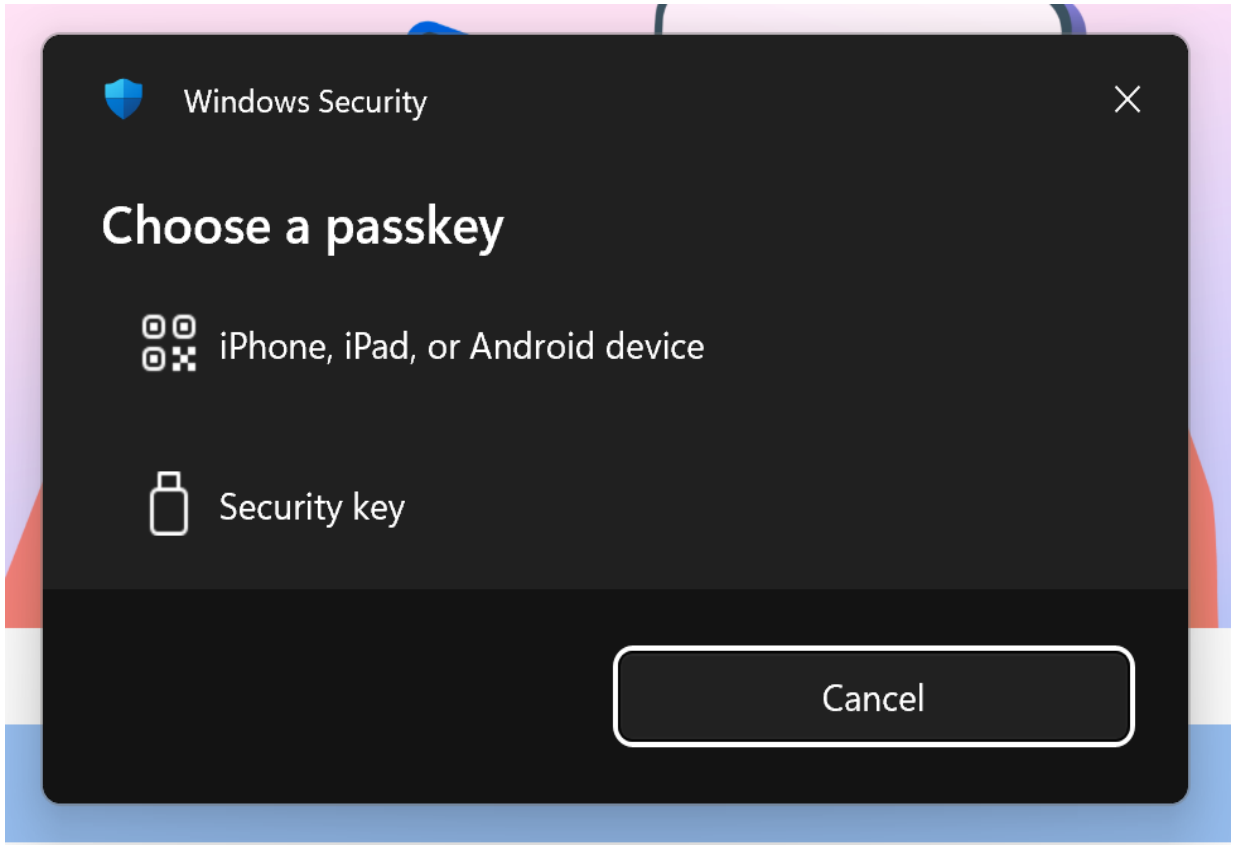
Step 2
Choose Continue



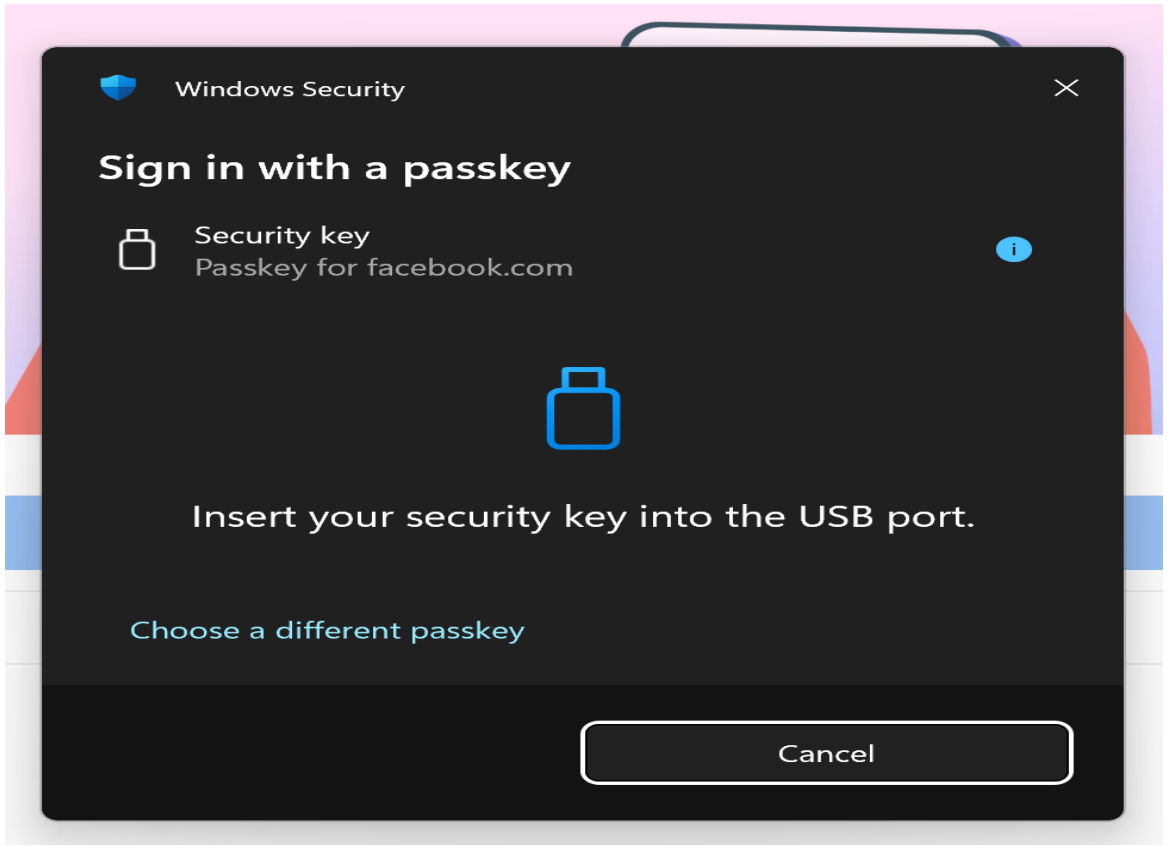
RESULT

Login Process (2-Factor Authenticaton)

Step 3
Choose Security Key



Step 4
Insert Security Device



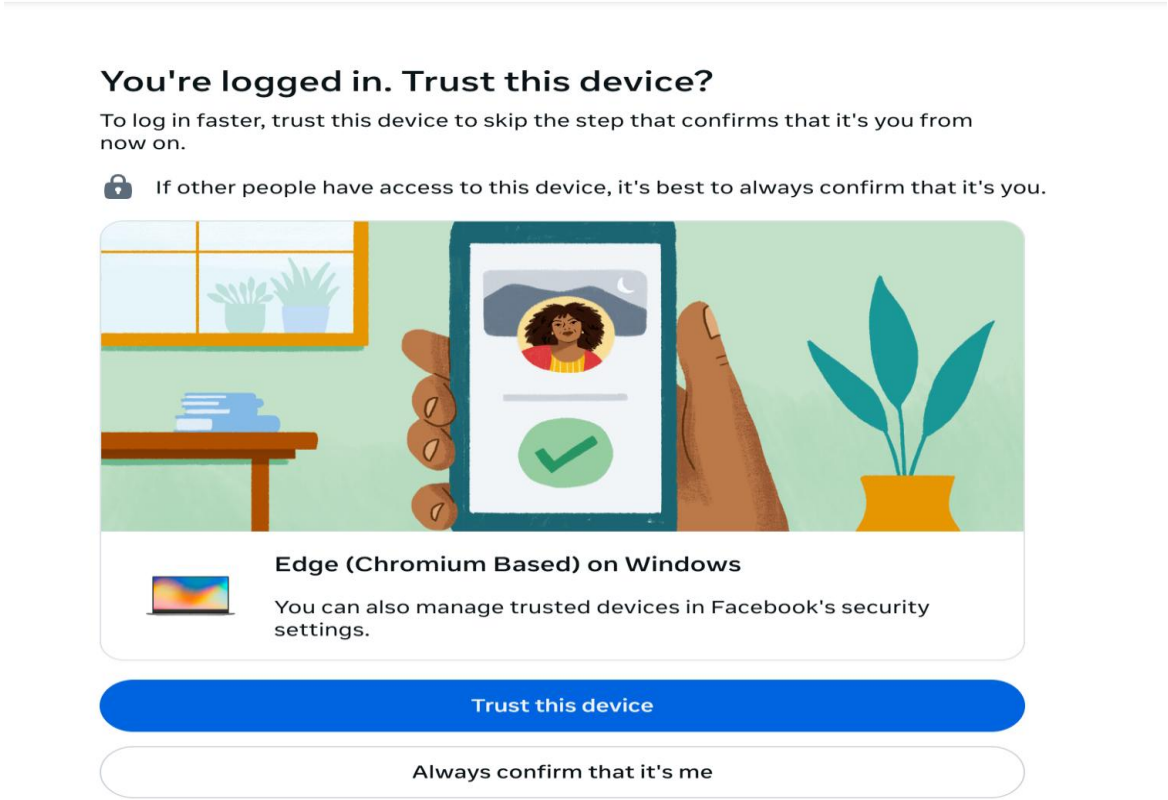
RESULT

Login Process (2-Factor Authenticaton)

Step 3
Enter Security Pin

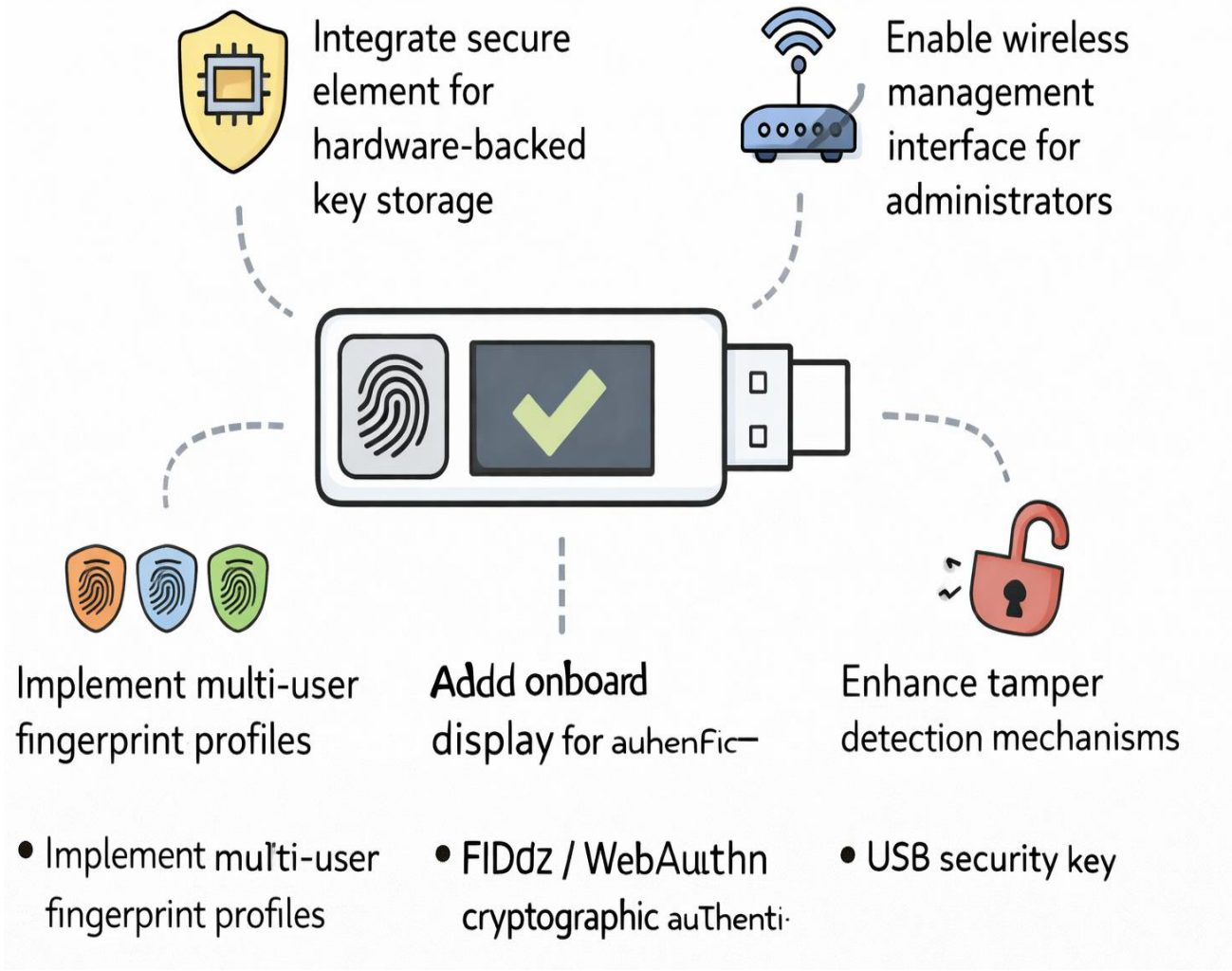


Step 4
Logged In



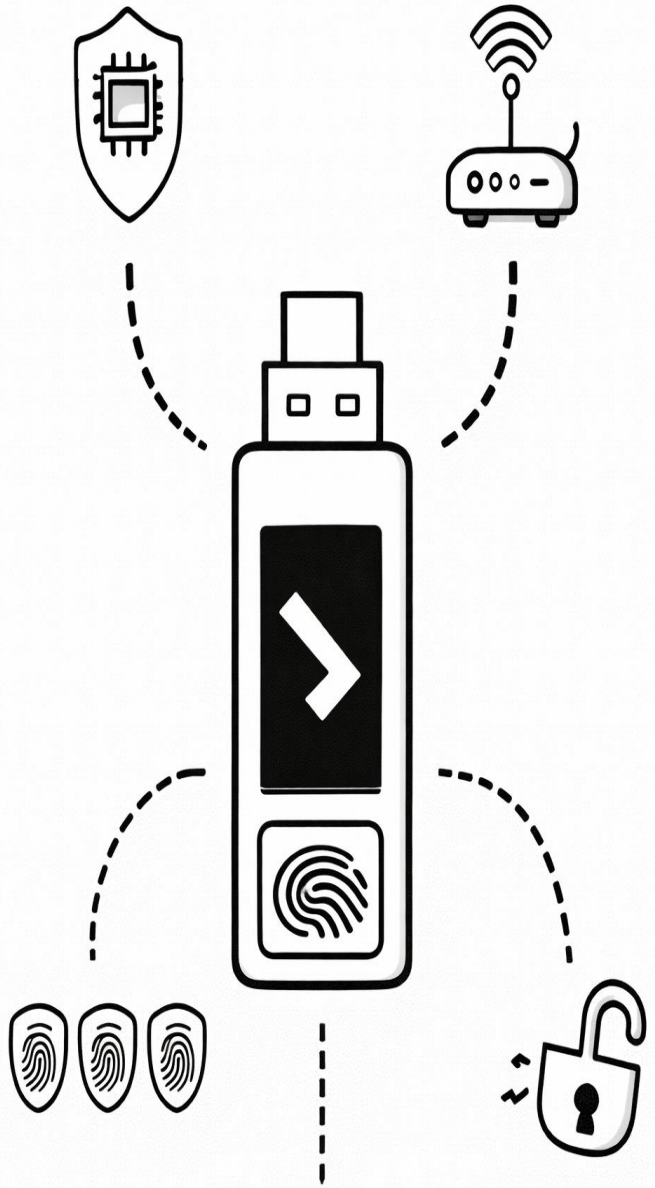
Future Enhancement

- Add onboard display for authentication status.
- Implement multi-user fingerprint profiles.
- Enable wireless management interface for administrators.
- Enhance tamper detection mechanisms.



Conclusion

- Biometric-gated hardware authentication system.
- Fingerprint + USB security key integration.
- Strong user identity assurance.
- High usability and convenience.
- Separate biometric & cryptographic processing.
- Layered, secure architecture design.
- Suitable for modern authentication needs.
- Scalable for future security enhancements.



References

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- https://gupea.ub.gu.se/server/api/core/bitstreams/
- 7fa3d00e-2ad8-4af3-8119-eddd895546ab/content
- https://www.researchgate.net/publication/387512723_Modified_USB_Security_Token_for_User_Authentication

Questions & Answers

Let’s explore your thoughts.



THANK YOU!

For your attention.

